

HIT137

Group Assignment 2 (20% Mark)

"assignment2.zip" contains all the files for this assignment.

You are required to create a GitHub repository and add all your group mates to it (make sure to keep it public, not private). You should do this before you start the assignment.

All the answers and contributions should be recorded in GitHub till you submit the assignment.

Submission Guidelines:

- Include your GitHub Repository link in a text file "github_link.txt".
- Zip all the programming files and outputs and "github_link.txt" and upload them to Learline.

Question 1

Create a program that reads the text file "raw_text.txt", encrypts its contents using a simple encryption method, and writes the encrypted text to a new file "encrypted_text.txt". Then create a function to decrypt the content and a function to verify the decryption was successful.

Requirements

The encryption should take two user inputs (shift1, shift2), and follow these rules:

- For lowercase letters:
 - If the letter is in the first half of the alphabet (a-m): shift forward by $\text{shift1} * \text{shift2}$ positions
 - If the letter is in the second half (n-z): shift backward by $\text{shift1} + \text{shift2}$ positions
- For uppercase letters:
 - If the letter is in the first half (A-M): shift backward by shift1 positions
 - If the letter is in the second half (N-Z): shift forward by shift2^2 positions (shift2 squared)
- Other characters:
 - Spaces, tabs, newlines, special characters, and numbers remain unchanged

Main Functions to Implement

Encryption function: Reads from "raw_text.txt" and writes encrypted content to "encrypted_text.txt".

Decryption function: Reads from "encrypted_text.txt" and writes the decrypted content to "decrypted_text.txt".

Verification function: Compares "raw_text.txt" with "decrypted_text.txt" and prints whether the decryption was successful or not.

Program Behaviour

When run, your program should automatically:

1. Prompt the user for shift1 and shift2 values
2. Encrypt the contents of "raw_text.txt"
3. Decrypt the encrypted file
4. Verify the decryption matches the original

Question 2

Write a program that reads mathematical expressions from a text file (one per line), evaluates each expression, and writes the results to an output file. Your solution should be built from plain functions, no classes. You should use recursive descent parsing, where each level of operator precedence is handled by its own function, and parenthesised sub-expressions are evaluated by recursing back to the top.

Requirements

- The evaluator must correctly handle the four basic operators (+, -, *, /), parentheses (nested to any depth), and unary negation (e.g., -5, --5, -(3+4)).
- Unary + is not supported and should produce an error.
- Unary negation may appear at the start of an expression, after an opening parenthesis, or after any operator (e.g., 3 * -2 is valid).
- Implicit multiplication is valid expression.

Output Format

Your program must produce a single output file called "output.txt" in the same directory as the input file. Each expression produces a four-line block (Input, Tree, Tokens, Result), with blocks separated by a blank line.

Tree

- A number literal is displayed as its formatted value.
- A binary operation is displayed as (op left right), the operator symbol comes first, followed by the left and right sub-trees, all separated by single spaces, enclosed in parentheses.
- A unary negation is displayed as (neg operand), using the word neg.
- Implicit multiplication appears in the tree as *.

Tokens

- Valid token types are: NUM (a numeric literal), OP (one of +, -, *, /), LPAREN (a (), RPAREN (a)), and END (end of input).
- Unary negation is not folded into the number token. The expression -5 produces the tokens [OP:-] [NUM:5] [END], not [NUM:-5] [END].

Output Formatting

- Input: line shows the original expression exactly as read from the file.
- Tree: line representing the parse tree, or ERROR.
- Tokens: line shows each token in the format [TYPE:value] separated by spaces, ending with [END], or ERROR.
- Result: line shows the computed value, or ERROR. If the result is a whole number (e.g., 8.0), display it without the decimal point (e.g., 8). Otherwise rounded to 4 decimal places.

Required Interface

Create a file "evaluator.py" that defines:

```
def evaluate_file(input_path: str) -> list[dict]:
```

- input_path is a string to the input text file.
- The function writes "output.txt" to the same directory as the input file.
- The function returns a list of dictionaries, one per expression, for example:

```
[  
    {"input": "3 + 5", "tree": "(+ 3 5)", "tokens": "[NUM:3] [OP:+] [NUM:5] ", "result":  
    8.0},  
    {"input": "3 @ 5", "tree": "ERROR", "tokens": "ERROR", "result": "ERROR"},  
]
```

The "result" value is a float on success or the string "ERROR" on failure. The "tree" and "tokens" value are strings.

A sample input file (sample_input.txt) and the expected output (sample_output.txt) are provided.